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Dynamic Black-Litterman

Anas Abdelhakmi, Andrew Lim

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The Black-Litterman model is a framework for incorporating forward-looking expert views in a portfolio optimization problem. Existing work focuses almost exclusively on single-period problems with the forecast horizon matching that of the investor. We consider a generalization where the investor trades dynamically and views can be over horizons that differ from the investor. By exploiting the underlying graphical structure relating the asset prices and views, we derive the conditional distribution of asset returns when the price process is geometric Brownian motion, and show that it can be written in terms of a multi-dimensional Brownian bridge. The components of the Brownian bridge are dependent one-dimensional Brownian bridges with hitting times that are determined by the statistics of the price process and views. The new price process is an affine factor model with the conditional log-price process playing the role of a vector of factors. We derive an explicit expression for the optimal dynamic investment policy and analyze the hedging demand

for changes in the new covariate. More generally, the paper shows that Bayesian graphical models are a natural framework for incorporating complex information structures in the Black-Litterman model. The connection between Brownian motion conditional on noisy observations of its terminal value and multi-dimensional Brownian bridge is novel and of independent interest.

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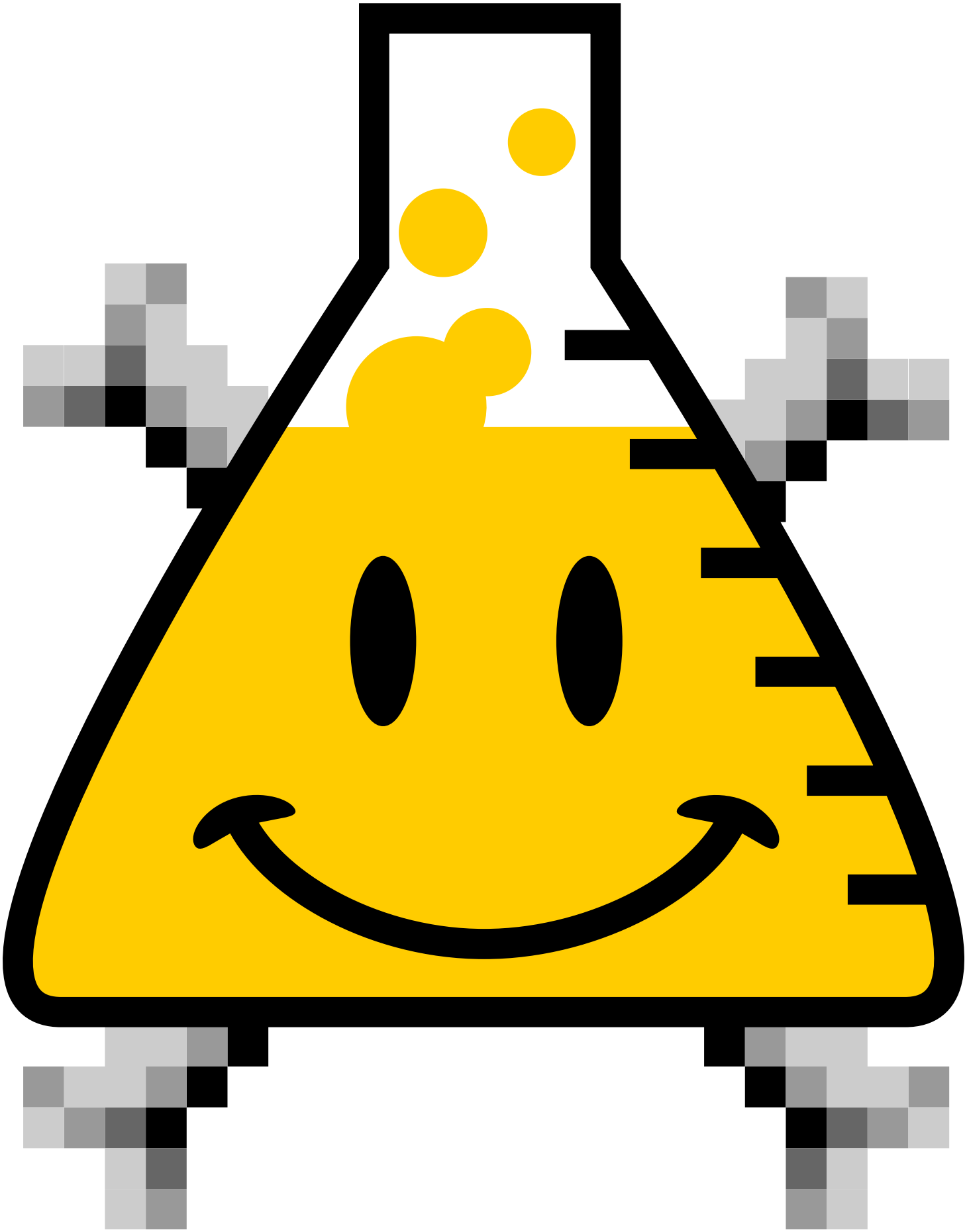

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